

■ CRIME INTELLIGENCE & ANALYTICAL PLATFORM

KSP Datathon 2026 — Hack2Skill

Track 2: AI-Driven Crime Analytics & Visualization Platform

Prize Pool	■10,00,000
Registration Deadline	July 19, 2026
Grand Finale	In-person, Bengaluru
Host	Karnataka State Police / SCRB

"Transform the Karnataka State Police's fragmented, Excel-based crime records into a real-time, AI-powered Strategic Intelligence Hub — enabling proactive policing, network analysis, and spatiotemporal crime visualization across all 31 districts."

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Re-architecting CIAP for Zoho Catalyst

The current CIAP architecture relies on generic cloud-native tools (Kubernetes, Keycloak, Kafka, MinIO, etc.). Based on the Datathon rules provided in the images, **deployment via Catalyst is mandatory**, and using third-party alternatives when a Catalyst service exists will negatively impact the submission.

This plan details the complete migration of our architecture to a 100% Zoho Catalyst-native stack.

User Review Required

[!IMPORTANT]

This is a major architectural overhaul. Please review the service mapping below to ensure you are comfortable using these Zoho Catalyst features during the hackathon.

Proposed Changes: Catalyst Service Mapping

We will update the `KSP_CIAP_Documentation.md` and `solution_architecture.md` (and regenerate the PDF) to reflect this new stack.

1. Frontend & API Routing

- **React SPA Hosting:** Moved from Nginx/K8s to **Catalyst Web Client Hosting**.
- **API Gateway & Routing:** Moved from Kong to **Catalyst API Gateway** (handling routing, throttling, and API auth).
- **User Auth / RBAC:** Moved from Keycloak to **Catalyst Authentication**.

2. Backend & Compute

- **Core Backend (FastAPI):** Moved from standard Docker to **Catalyst AppSail (managed runtime)** or **Catalyst Serverless (Functions)** depending on endpoint complexity.
- **Cron Jobs / ETL (Airflow replacement):** Replaced with **Catalyst Cron / Job Scheduling**.
- **Data Pipelines / Orchestration:** Replaced with **Catalyst Circuits** for multi-step workflows.

3. Data & Storage

- **Relational DB (FIRs, Suspects):** Moved from PostgreSQL to **Catalyst Data Store**.
- **File Storage (Images, Docs):** Moved from MinIO to **Catalyst Stratus** (Object Storage).
- **Cache:** Moved from Redis to **Catalyst Cache**.

■ **Event Bus / Real-time (Kafka replacement):** Replaced with **Catalyst Signals & Event Functions** (e.g., triggering anomaly detection when a new FIR is inserted).

4. AI & ML (The biggest upgrades)

■ **Text Analytics / NER (spaCy replacement):** Replaced with **Catalyst Zia Services (Text Analytics)** to extract entities from FIR narratives natively.

■ **LLM & RAG (OpenAI replacement):** Replaced with **Catalyst QuickML (LLM Serving, RAG)** for the natural language querying assistant.

■ **Predictive Risk Scoring:** Can utilize **Catalyst Zia AutoML** for tabular model training based on historical crime data.

■ **Custom ML (ST-DBSCAN):** Hosted via **Catalyst AppSail (custom OCI runtime)** since it requires specific Python spatial libraries.

5. Utilities

■ **Report Generation:** Moved from React-PDF to **Catalyst SmartBrowz** for generating PDF reports of crime dashboards.

■ **Alerts:** Replaced WebSockets with **Catalyst Push Notifications**.

Open Questions

[!WARNING]

Graph & Spatial Data Support: Catalyst Data Store is a standard relational store.

1. Do you want to host **Neo4j** inside a **Catalyst AppSail (custom OCI runtime)** container, or should we drop the Graph DB requirement and simplify network analysis using recursive SQL in Catalyst Data Store?

2. For geospatial hotspots (ST-DBSCAN), we will manage the spatial calculations in Python on AppSail rather than relying on PostGIS. Is this acceptable?

Verification Plan

1. Update **KSP_CIAP_Documentation.md** with the new Catalyst-native architecture diagrams and tech stack tables.
2. Update **solution_architecture.md**.
3. Re-run the **generate_pdf.py** script to update **KSP_CIAP_Documentation.pdf** in **D:\Projects\Lumina**.
4. Ensure all 26 capabilities from the provided image are mapped correctly if used.